

基础

```
public String toString() {  
    return "...";  
}  
public ...() {  
    super("...");  
}
```

Array/ArrayList

数组的属性：List.Length

GUI

```
JFrame myJFrame = new JFrame();  
JButton myJButton = new JButton("context");  
MyFrame.getContentPane().add(myJButton);  
myJButton.addActionListener(new ActionListener() {  
    public void actionPerformed (ActionEvent e) {  
        myJButton.setText("newtext");  
    }  
});
```

IO

```
import java.io.*;  
读：  
File myFile = new File("filename.txt")  
//当题目中要求新建文件时写这段;  
if (myFile.exists()) {  
    System.out.println("File exists");  
    System.exit();  
}  
try {  
    FileWriter output = new FileWriter(myFile);  
    output.write(String + "\n");  
    output.close();  
}  
catch (IOException ex) { ex.printStackTrace(); }
```

写:

```
try {
    FileReader file = new FileReader(args[0]);
    BufferedReader in = new BufferedReader(file);
    int numLines = Integer.parseInt(args[1]);
    for (int lineCt = 0; lineCt < numLines; lineCt++) { //while (in.readLine() != null)
        String line = in.readLine();
        System.out.println(line);
    }
}
catch (Exception e) { System.out.println("Error: " + e);}
//可先 catch FileNotFoundException 以及 IOException
Finally {
    in.close();
    file.close();
}
```

Exception

```
if (...)
    throws new ...Exception();
```

Exception in thread "main" java.lang.IndexOutOfBoundsException: Index a out of bounds for length b

Methods

GUI (Swing/AWT)

Jframe

```
setTitle(String)
setSize(int, int)
setVisible(boolean)
add(Component)
```

JButton

```
setText(String)
addActionListener(ActionListener)
```

JLabel

```
setText(String)
setHorizontalAlignment(int)
setVerticalAlignment(int)
```

JTextField

setText(String)
getText()
addActionListener(ActionListener)
setColumns(int)

IO

File

exists()
isFile()
isDirectory()

FileInputStream/FileOutputStream

read()
write(int)
close()

BufferedReader/BufferedWriter

readLine()
read()
write(String)
newLine()
close()

Exception

Exception 类

toString()

常见异常类型

NullPointerException
ArrayIndexOutOfBoundsException
FileNotFoundException
IOException

String

基本方法

length()
charAt(int)
substring(int)
substring(int, int)

indexOf(String / char)
lastIndexOf(String)

比较方法

equals(Object)

修改方法

toLowerCase()
toUpperCase()

Math

基本运算

abs(double)
max(double, double)
min(double, double)

取整方法

round(double / float)

随机数和常数

random() / PI / E

ArrayList

添加元素

add(Object) / add(int, Object)

删除元素

remove(int) / remove(Object)

查询元素

get(int)
indexOf(Object)
lastIndexOf(Object)

修改元素

set(int, Object)

大小和状态

size()
isEmpty()

转换方法

toArray()